

ROHITH RAJ BAGGAM

Backend Engineer | Technical Lead | Solution Architect

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Professional Summary

Architecture-first Backend Engineer and Technical Lead (3 years, Aptagrim) who treats backend as system design, not just coding — and is framework- and language-agnostic across stacks. Led backend teams on six production platforms: multi-tenant microservices, real-time systems, database engineering, a 20-year legacy MSSQL modernization, and network-telemetry research. Python-first (Django, Django REST Framework, FastAPI) with Java (Spring Boot, Quarkus, Hibernate) through independent projects; author of two published PyPI packages. Seeking larger-scale systems and strong senior engineering mentorship.

Technical Skills

Languages & Frameworks: Python, Java, SQL; Django, Django REST Framework (DRF), FastAPI, Flask, Spring Boot, Quarkus, Hibernate/Panache, Jakarta EE, Django Channels

Databases: PostgreSQL, MySQL, MSSQL, Oracle, MongoDB, Neo4j, SQLite; JDBC, PgBouncer connection pooling

Async & Real-time: Celery, Celery Beat, Redis, WebSockets, ASGI, Daphne, Uvicorn

Networking & Telemetry: SNMPv1, SNMPv2c, SNMPv3, WMI, SSH, ICMP, NetFlow, IPFIX, Syslog, Windows Event Logs (pysnmp, pywinrm, paramiko)

Integrations: LiveKit, Google Maps (Geocoding & Address Validation), Mappls / MapMyIndia (eLoc), OSRM, OnlyOffice, Mailcow (IMAP/SMTP), Firebase FCM, FFmpeg

Cloud, Security & DevOps: E2E Compute Nodes & EOS, Azure Blob Storage, MinIO/S3; JWT (HS256/RS256), AES-GCM, HMAC blind-index, Fernet, OTP, RBAC; Docker, GraalVM native image, Nginx, Gunicorn, GitLab CI, GitHub Actions, pre-commit, SonarQube, Postman, DBML; pandas, numpy, openpyxl

Professional Experience

Aptagrim Limited — Backend Engineer → Technical Lead

Hyderabad, India | Jul 2023 – Present

PsyHire v2 — ATS / Recruitment Platform

Solution Architect | May 2026 – Present

- Architected a multi-tenant recruitment platform as three Django microservices (control-plane/admin, auth, application) — each tenant gets its own PostgreSQL database and Redis logical DB, resolved at runtime through a Redis-backed tenant registry on a shared platform layer (ContextVar tenant routing, PyJWT auth, generic DRF framework).
- Designed cross-service tenant provisioning with service-token migration orchestration and transactional rollback across databases and Redis, so a failed onboarding never leaves orphaned state.
- Integrated LiveKit interviews with Room Composite Egress recording, external AI APIs for resume parsing/scoring and JD generation, Celery-driven hiring-pipeline automation, and Django Channels websockets for live pipeline and notification updates.

Ping — Enterprise Communication & Collaboration Platform

Technical Lead & Core Backend Developer

- Led backend and solution architecture for a self-hosted Slack/Zoom-style platform (chat, files, calendar, meetings, mail, notifications); designed a split Redis topology isolating user/session and encrypted-mailbox state from the Celery broker, Channels layer, pub/sub and distributed locks.
- Built secure real-time messaging with AES-GCM encrypted content, HMAC blind-index search over encrypted messages, presigned private object storage, and OnlyOffice document editing via JWT-signed editor sessions with callback validation.
- Integrated Mailcow IMAP/SMTP mailboxes, LiveKit meetings/recordings, Firebase FCM push notifications, and an FFmpeg/Pillow Celery media-compression pipeline guarded by Redis distributed locks.

AIN — Network Monitoring & Discovery Platform

Database Engineer & Research Analyst

- Engineered multi-tenant async FastAPI + SQLAlchemy backends with runtime per-tenant database routing and safe, non-destructive schema-sync migrations for a SolarWinds-class monitoring platform.
- Built time-series monitoring analytics over PostgreSQL — CPU, RAM, disk and interface metrics, NetFlow/IPFIX flow analysis, Syslog and Windows Event Log telemetry — using tuned raw SQL with dynamic aggregation windows.
- Researched and implemented protocol-based device discovery across SNMPv1/v2c, SNMPv3, WMI, SSH and ICMP, normalizing heterogeneous device facts into a protocol-agnostic inventory schema; hardened access with JWT middleware, SNMPv3 auth/privacy and tenant isolation.

Sigma — BFSI Loan Recovery Platform

Technical Lead | Jul 2025 – Jan 2026

- Architected and deployed the full backend stack on E2E Compute Nodes + EOS object storage for a field-agent loan-recovery platform serving managers and field officers across a geographic hierarchy.

- Built real-time field-officer tracking and route optimization with the OSRM routing engine and WebSocket live-location streaming, plus a rule-based auto-assignment engine enforcing territory, capacity, proximity and performance constraints.
- Integrated Google Maps geocoding/address-validation, Mappls eLoc and LiveKit ASGI video calling; delivered bulk allocation/referral file ETL with websocket progress reporting and ORM-aggregated web/mobile dashboards.

TruOperate — Industrial Operations Management System

Backend Technical Lead | Feb 2024 – Jul 2025

- Led zero-downtime modernization of a 20-year-old Java platform to Python/Django while preserving the production Microsoft SQL Server schema through manual field-by-field model mapping and SQL-based migration bridges.
- Built security-group role-based access control, operator-round, log-book, equipment and alert APIs with metadata-driven permission automation, plus TruTech service APIs for mobile/field sync.
- Integrated JasperReports through a Java-to-Python bridge for enterprise report generation, Excel export workflows, and Celery/Celery Beat scheduled processing over the legacy MSSQL estate.

PsyHire v1 — ATS / HR-Tech Platform

Core Backend Developer | Aug 2023 – Feb 2024

- Designed scalable DRF REST APIs with asynchronous Celery + Redis processing for bulk email campaigns, reminder triggers and ML model inference.
- Worked across relational (PostgreSQL) and NoSQL (MongoDB) stores and led media-storage migration from Azure Blob to E2E/GCP object storage.

Open-Source & Personal Projects

OneDock Studio — Native Backend Engine for a Cross-Database IDE

Author | GitHub

- Built the native backend engine of a cross-engine database workbench/IDE in Java 21 + Quarkus — designed to compile to a GraalVM native image and embed inside the desktop app as a localhost sidecar owning all database and OS-level interaction; Hibernate ORM with Panache over SQLite for engine metadata, and raw JDBC across PostgreSQL, MySQL, MSSQL and Oracle for target databases, with per-engine differences isolated behind a strategy/factory layer.
- Implemented per-profile RS256 JWT authentication, JDBC schema introspection with automated ER-diagram and FK-dependency generation, a safe ad-hoc SQL execution engine with row caps/timeouts/truncation detection, and a database-native audit-trigger generator (twin-table + trigger strategies).

django-data-seed — Open-Source PyPI Package

Author | PyPI | GitHub

- Zero-config Django test-data generator that fills an entire database from one command — inspects models, resolves table order, and generates realistic, relationship-aware, constraint-respecting values, eliminating hand-written factories and fixtures.
- Bulk-inserts hundreds of thousands of rows in seconds with reproducible seeded output and a pre-flight check that halts on unapplied migrations or model drift; CI-verified across Python 3.10–3.14 × Django 4.0–6.0 on SQLite, PostgreSQL and MySQL.

django-perfy — Open-Source PyPI Package

Author | PyPI | GitHub

- API, WebSocket and server performance monitoring for Django: middleware captures endpoint latency, DB query count and DB time; a Channels mixin times WebSocket events; Celery tasks snapshot CPU/memory/Redis/Postgres resources.
- Rolls telemetry into minute/hour summaries with a staff dashboard and downloadable/emailable PDF reports; supports a secondary telemetry database via a bundled DB router, with sampling and PII-safe redaction.

Quarkus Blog Backend — Java Learning Project

Solo Developer | GitHub

- REST backend exploring the full Quarkus stack: RS256 JWT auth with a custom request filter, Hibernate ORM with Panache, Flyway migrations, pagination/filtering, standardized response wrappers, global exception mapping and OpenAPI docs in a layered Resource/Service/Repository architecture.

Education

B.E. Computer Science Engineering — Chandigarh University, Punjab

2019 – 2023

CGPA: 7.0/10

Certifications

Google Cloud Platform — Production Deployment • Neo4j Graph Data Science & Cypher • Code Quality (SonarQube, pre-commit, PEP-8)

Professional Strengths

Backend team leadership • Solution architecture • Product-management mindset • On-time delivery & ownership • Fast learner • Optimization-driven